Adaptive Smooth Adversarial Learning for Cross-Domain Hyperspectral Image Classification

Rong Liu, *Member, IEEE*, Mengqing Zhou, Jiaqi Yang, and Junjue Wang

***Abstract*—Cross-domain classification has emerged as a widely employed method in remote sensing due to the challenges associated with obtaining labels for hyperspectral image (HSI). To address the impact of inter domain differences on classification performance, domain-adaptive deep learning (DL) methods have proven effective in recent years. However, many existing methods struggle to effectively integrate high-dimensional information in hyperspectral cross-scene classification, are insufficient for interclass discriminability, and are sensitive to small perturbations. These limitations hinder the accurate extraction of meaningful features across different domains. To tackle these issues, we propose a cross-scene HSI classification method named adaptive smooth adversarial learning (ASAL). This approach utilizes a dynamic attention mechanism to capture region semantic information, where the spectral attention deformable block (SADB) is designed to achieve dense pixel-level attention. Starting from the concept of class confusion, combining category minimization with multiscale analysis, we design entropy-weighted minimum-class confusion (EMC) loss, which improves the ability to capture interclass relationships. The sharpness of the loss function is also reduced using a double smoothing regularization strategy, which enhances the generalization ability of the model. Experimental results show the proposed method improves accuracy by about 3.00% or higher over state-of-the-art methods across four cross-scene HSI benchmarks. The code for the proposed network will be available at https://github.com/liurongwhm**

***Index Terms*—Adversarial learning, cross-domain classification, deep learning (DL), hyperspectral image (HSI) classification, unsupervised domain adaptation (DA).**

I. Introduction

HYPERSPECTRAL images (HSIs) consist of continuous, narrow spectral bands containing rich spectral and spatial information [1]. They have been widely applied in various fields, including environmental protection [2], [3], agricultural monitoring [4], [5], and other fields [6], [7], [8]. The rapid advancement of deep learning (DL) has significantly expanded the use of DL methods in HSI classification [9], [10], [11], achieving promising results when sufficient labeled samples are available. However, most existing methods require large amounts of labeled data and assume a consistent data distribution [12]. Therefore, a key research challenge in HSI classification is the development of models with strong generalization capabilities to reduce labeling costs [13].

Transfer learning [14], [15] facilitates the transfer of information between different HSI scenes by leveraging knowledge or models from a source domain (SD) to improve classification performance in a target domain (TD). Cross-domain classification [16], [17], [18], a specific application of transfer learning, involves classifying the TD using labeled data from the SD, typically in scenarios where the TD has little to no labeled.

However, factors such as acquisition time [19], geographical location, and imaging conditions introduce significant differences in data distribution between domains. As a result, acquiring domain-invariant knowledge remains a challenging task [20], [21]. Inspired by classical machine learning methods, several domain adaptation (DA) methods for HSI classification have been proposed [22], [23]. Multisource iterative DA (MSIDA) [24] enhances cross-domain classification by selecting the top-K SDs that based on their similarity and complementarity with the TD and iteratively learning TD-specific features from them. Persello and Bruzzone [25] measured the distance between the conditional distributions of the SD and TD in the reproducing kernel Hilbert space. While these methods have shown some success in cross-domain classification, they face limitations such as insufficient feature representation, poor adaptability to cross-domain distribution shifts, and limited robustness and generalization. These challenges limit their effectiveness in meeting the requirements of cross-domain tasks.

Compared with traditional shallow models [26], [27], deep networks can effectively utilize advanced nonlinear features of images [28], [29], [30], [31]. DL-based methods reduce distribution differences by aligning feature spaces, enabling SD models to generalize well to TDs. Feature space alignment can be categorized into statistical feature alignment and adversarial alignment, both addressing cross-domain challenges.

Statistical alignment methods minimize distribution differences by utilizing the statistical properties of features and matching distributions based on specific metrics. For example, maximum mean discrepancy (MMD) [32] aligns mean values by incorporating an MMD term into the loss function, ensuring the average features of both domains are as close as pos-

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sible. Similarly, correlation alignment (CORAL) [33] aligns covariance matrices across domains, reducing discrepancies in feature distributions.

In contrast, adversarial alignment employs deep adversarial networks for feature alignment. Dynamic adversarial adaptive network (DAAN) [34] employs a domain discriminator to differentiate between source and target features, optimizing the feature extractor via backpropagation until the source and target features become indistinguishable. The unsupervised joint adversarial DA (UJADA) [35] enhances domain-level and class-level feature alignment by incorporating a classifier discrepancy metric, which quantifies probabilistic differences in label predictions on target data, enabling more precise feature alignment at a finer granularity.

Although the above DL-based approaches have demonstrated improved performance, several challenges remain in advancing cross-domain classification for HSIs.

First, HSI data often exhibit spatial shape variations in targets and complex background interference, making it challenging to fine-tune feature extraction in specific regions. Effective classification requires identifying features that not only distinguish land cover classes but also ensure domain invariance between the source and TDs [36], [37]. Spatial–spectral–semantic cross-domain few-shot learning (S3CFSL) [38] combines spatial and spectral dual channels with a cross-spatial–spectral transformer to extract fused spatial–spectral features. Wang et al. [39] introduced a weak parameter sharing mechanism to balance domain-invariant and domain-specific features. While convolutional neural networks (CNNs) are used to extract high-level features suitable for pattern recognition, their limited receptive field constrains the ability to capture contextual information beyond local spatial boundaries [40], [41], [42]. Integrating multidimensional information during feature extraction becomes challenging when spatial and spectral attention are treated as independent processes. The complex interactions between spatial and spectral dimensions in HSI data often result in certain spectral features being discriminative only in specific spatial patterns. To address this, we propose introducing a pixel-level attention mechanism. This mechanism would enable the model to dynamically adjust the attention paid to each pixel in the global context, enhancing regional semantic detail and capturing the intricate interactions between spatial and spectral features effectively.

Second, enhancing interclass discriminability and reducing class confusion are critical challenges in HSI classification. Significant progress has been made in migratable feature learning through adversarial DA and cross-domain knowledge transfer, which has improved the effective transfer of critical features [43]. Kurmi et al. [44] incorporated probabilistic determinism into classifier training, employing estimation techniques to emphasize key regions and reduce foreground interference during classification adaptation. Unsupervised heterogeneous DA CycleGan (UHDAC) [45] utilizes a CycleGAN framework to capture similarity-independent transferable features, integrating the CORAL loss function to ensure seamless feature transfer between the SD and TD through cycle consistency. However, in multiclassification tasks, even when adversarial training brings the discriminators to a state of complete confusion, it remains difficult to ensure sufficient alignment between source and TD distributions [46]. This misalignment weakens intercategory discriminability and limits its classification accuracy. Guo et al. [47] highlighted that deep neural networks (DNNs) often exhibit overconfidence in specific categories, which impedes accurate reasoning and identification of interclass ambiguities. To address this, introducing targeted constraints or refined loss functions during adversarial training can enhance the distinction between category boundaries while mitigating the network's tendency toward overconfidence in certain categories. This approach fosters a more consistent and transferable feature distribution between the source and TDs.

Third, the high-dimensional features and spectral redundancy of HSI increase the risk of overfitting to SD details and reduce robustness during domain transfer [48], [49], [50]. Lee and Kwon [51] proposed a self-supervised method to learn representations from unlabeled HSIs to improve adaptation to the TD. This approach ensures decision boundaries in the TD align with low-density regions. However, such methods often focus on feature space alignment or adversarial training, neglecting the model's sensitivity to small perturbations in input data. HSI data frequently contains noise or spectral variability, which, if unaddressed, can cause large fluctuations in classification outcomes. To improve generalization, minimizing both the training loss and the sharpness of the loss function could help mitigate the impact of these perturbations.

To address the three challenges mentioned above, we propose a novel method for HSI classification. The main contributions of this study are as follows.

1) To overcome the limitations in integrating high-dimensional information during cross-domain classification of HSIs, we introduce the spectral attention deformable block (SADB). The proposed model comprises three key components: gated attention, deformable convolution (DC), and spectral normalization. The gated attention mechanism allows the model to automatically focus on critical regions, improving its sensitivity to essential features. DC optimizes the ability to capture relevant image regions by dynamically adjusting the convolution kernel's position. In addition, spectral dimension reduction and spectral normalization effectively manage the complexity of HSI data, improving feature extraction and classification performance.
2) To address the instability of cross-domain feature transfer and the insufficient category differentiation in adversarial training, we propose an entropy-weighted minimum-class confusion (EMC) loss integrated with multiscale feature analysis. This approach mitigates the model's overconfidence in specific categories by employing dynamic temperature scaling, while an entropy-weighting mechanism enhances sensitivity to high-entropy samples, allowing the model to focus more effectively on uncertain and critical samples. By incorporating multiscale feature analysis, our method captures category distribution characteristics and interclass relationships across different scales, improving the sta-

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bility of cross-domain feature alignment and enhancing category differentiation.

3) To address the overfitting caused by high-dimensional features and spectral redundancy in HSI and improve robustness to the TD, we propose a strategy to optimize convergence to a smoothed minimum. We propose label-smoothing DA (LSDA), which integrates smoothing domain adversarial training (SDAT) and label smoothing to jointly regularize the model in both feature and label spaces. This approach enhances noise resistance and stability in hyperspectral cross-domain classification. By implementing smooth alignment of feature distributions across the SD and TD, the interdomain discrepancies are effectively suppressed, while the overfitting of TD features is prevented.

II. Related Works

A. CNN-Based Methods—CNN Attention Mechanism

The CNN attention mechanism is developed to eliminate redundant information flowing through the neural network while addressing the computational load problem. The attention mechanism can be represented as

$$\text{Attention} = f\left(g\left(x\right), x\right) \tag{1}$$

where $g(x)$ denotes the generation of attention, which refers to the process of identifying discriminative regions. $f(g(x), x)$ represents the processing of the input x based on the attention $g(x)$, enabling the model to focus on key regions and extract relevant information.

CNN attention mechanisms can be broadly categorized into six classes [52]: channel attention, spatial attention, temporal attention, branching attention, channel–spatial attention, and spatiotemporal attention. Each of these mechanisms can be represented by (1). This study specifically focuses on channel attention, spatial attention, and channel–spatial attention.

Channel attention focuses on adaptively adjusting the weights of different channels, as each channel typically corresponds to distinct objects. SENet [53] introduced channel attention by utilizing global pooling and fully connected layers to capture interchannel relationships. In contrast, spatial attention identifies relevant spatial regions, dynamically selecting critical objects and areas. The Residual Attention Network [54] combines channel and spatial attention, producing 3-D attention maps (height, width, and channel). In scene segmentation, the dual attention network (DANet) [55] employs self-attention to capture global information by integrating positional and channel attention modules in parallel, with the final features fused from both branches.

Overall, existing methods need to effectively capture channel, spatial, and global dependencies, as these are crucial for understanding contextual information. However, dense attention mechanisms often lead to increased computational overhead, posing significant challenges. We want to be able to obtain pixel-level holistic attention that allows the model to focus on relevant information while maintaining the simplicity and efficiency of the CNN architecture.

B. Smoothness of the Loss Function

Modern DL models typically contain a large number of parameters, making them susceptible to overfitting, where they perform excellently on training data but struggle to generalize to test data [56]. Minimizing the training loss, such as cross entropy, is insufficient for finding model parameters that generalize effectively, especially when the loss function contains multiple local and global minima. The geometry of the loss function is closely linked to generalization ability, with flat minima—where the loss function is less sensitive to small parameter changes—corresponding to better generalization. In contrast, steep minima, where small parameter changes lead to a sharp increase in loss, are typically associated with poorer generalization.

Sharpness-aware minimization (SAM) [57] leverages the geometry of the loss function to enhance model generalization by minimizing both the loss value and the curvature of the loss function. SAM is formulated as a min–max optimization problem, where the goal is to minimize the maximum loss increase within a specified range of perturbations

$$\min_w \max_{\|\epsilon\| \le \rho} \mathcal{L}\left(w + \epsilon\right) \tag{2}$$

where w represents the model parameters, $\mathcal{L}(w)$ is the loss function, and ϵ is the perturbation applied to the parameters, constrained to a magnitude within a radius ρ.

To efficiently address the min–max optimization problem, SAM employs a first-order Taylor expansion to approximate the inner maximization component $\mathcal{L}(w + \epsilon)$

$$\max_{\|\epsilon\| \le \rho} L\left(w + \epsilon\right) \approx L\left(w\right) + \epsilon^{\mathrm{T}} \nabla_w L\left(w\right). \tag{3}$$

In general, only the first-order term is considered, resulting in the following approximation:

$$\max_{\|\epsilon\| \le \rho} L\left(w + \epsilon\right) \approx \rho \|\nabla_w \mathcal{L}\left(w\right)\|_2 \tag{4}$$

where $\|\nabla_w \mathcal{L}(w)\|_2$ is the Euclidean norm of the gradient.

Under this approximation, the optimization task has an analytical solution

$$\epsilon^*\left(w\right) = \rho \frac{\nabla_w \mathcal{L}\left(w\right)}{\|\nabla_w \mathcal{L}\left(w\right)\|}. \tag{5}$$

Once the optimal perturbation $\epsilon^*(w)$ is obtained, the parameters can be updated with $w + \epsilon^*(w)$

$$\nabla_w L_{\mathrm{SAM}}\left(w\right) \approx \nabla_w L\left(w + \epsilon^*\left(w\right)\right). \tag{6}$$

SAM ensures that the updated parameters reside in flatter regions of the loss landscape, thereby mitigating sharp increases in loss due to localized steep gradients. In hyperspectral cross-domain classification, where significant differences in data distributions often exist between the SD and TD, enhancing the model's generalization is critical. SAM's loss optimization strategy can be effectively leveraged to address these challenges.

III. Methodology

A. Proposed Framework

The proposed DA framework is developed for HSI classification. The flowchart of the proposed framework is shown

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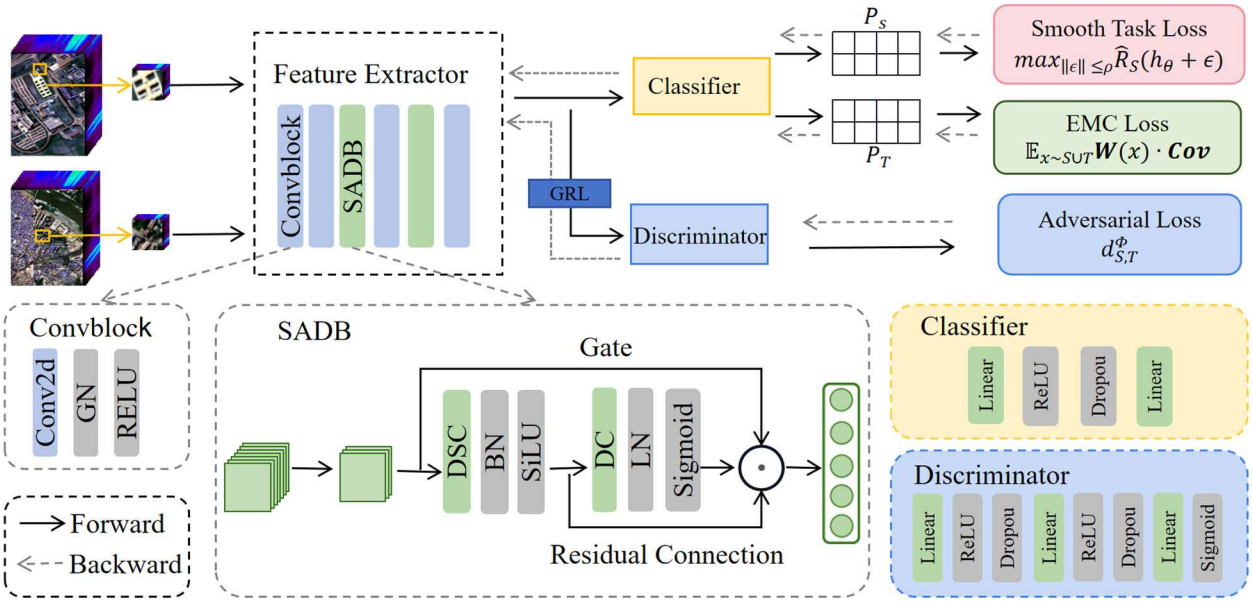


Fig. 1. Flowchart of the proposed framework. GRL stands for gradient reversal layer, and EMC Loss refers to entropy-weighted minimum-class confusion loss.

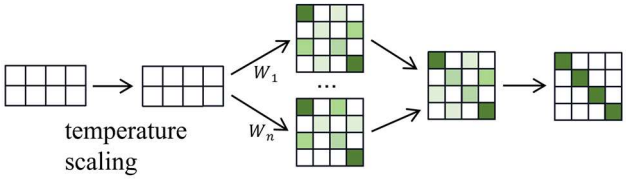


Fig. 2. Flowchart of the EMC Loss.

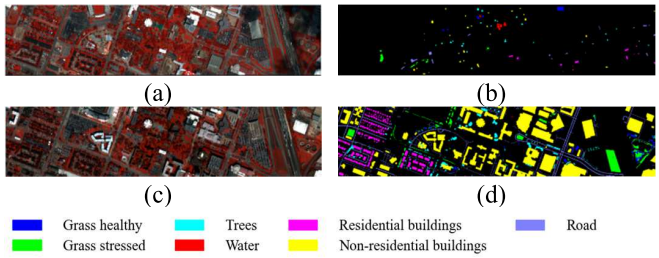


Fig. 3. Houston dataset. (a) Houston 2013 false-color image ([R, G, B]: bands [30, 15, 10]). (b) Houston 2013 ground-truth map. (c) Houston 2018 false-color image ([R, G, B]: bands [30, 15, 10]). (d) Houston 2018 ground-truth map.

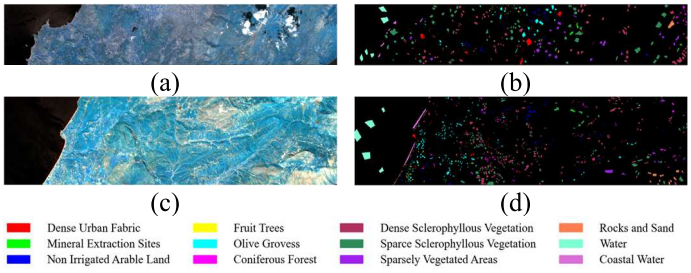


Fig. 4. HyRANK dataset. (a) Dioni false-color image ([R, G, B]: bands [20, 30, 40]). (b) Dioni ground-truth map. (c) Loukia false-color image ([R, G, B]: bands [20, 30, 40]). (d) Loukia ground-truth map.

in Fig. 1. Assume that $X = \{x_i\}_{i=1}^{N} \in R^d$ represents the data from the SD, and $Y = \{y_i\}_{i=1}^{N}$ demotes the corresponding class labels. Here, d represents the data dimensionality, and N denotes the number of source samples. The framework consists of three main components: a feature extractor, a classifier, and a discriminator. A spatial patch of size $p \times p \times d$ is selected from the HSI in the SD and sent to the feature extractor to derive metastable features. These features are classified to obtain category vectors, which guide the generator and discriminator, smooth the domain adversarial process, and enhance the network's discriminative capability. This collaborative processing enables the network to efficiently learn cross-scene invariant features.

This approach selects spatial patches from both the SD and TD in HSI, feeding them into a feature extractor that combines an attention mechanism with a CNN to capture relocatable features. These features are then classified to generate category vectors, while a generator and a discriminator collaboratively align outputs by minimizing the domain gap. Adversarial training enhances the generator's ability to produce indistinguishable features, effectively reducing interdomain differences. Simultaneously, the smooth domain adversarial process reinforces network stability and generalizability by minimizing the smoothing term of the task loss. Through stochastic gradient descent (SGD), the network achieves a smoother loss function, enhancing cross-scene invariant feature learning. This collaborative process improves adaptability and classification accuracy across diverse domains.

B. Spectral Attention Deformable Block

To efficiently and accurately capture invariant features in hyperspectral cross-domain scenes, we designed a feature

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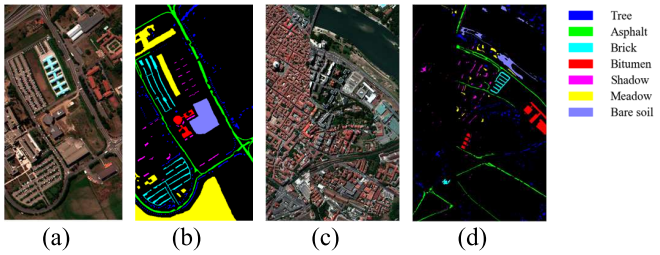


Fig. 5. Pavia image. (a) UP false-color image ([R, G, B]: bands [60, 30, 20]). (b) UP ground-truth map. (c) PC false-color image ([R, G, B]: bands [60, 30, 20]). (d) PC ground-truth map.

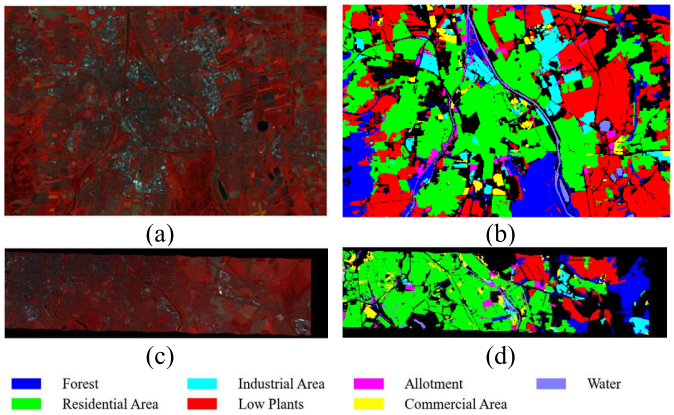


Fig. 6. C2Seg-AB image. (a) Augsburg false-color image ([R, G, B]: bands [70, 30, 20]). (b) Augsburg ground-truth map. (c) Berlin false-color image ([R, G, B]: bands [70, 30, 20]). (d) Berlin ground-truth map.

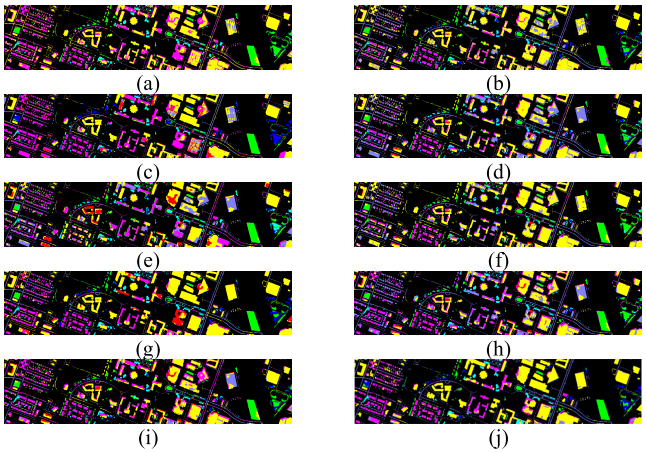


Fig. 7. Classification maps of different methods for Houston 2018. (a) DAAN. (b) DSAN. (c) TVT. (d) DDAN-JCA. (e) SDEnet. (f) S2ECNet. (g) TSTNet. (h) LLURNet. (i) S2AMNet. (j) ASAL.

extractor that directly integrates an attention mechanism into the CNN, optimized for the high-dimensional nature of HSI data. This approach effectively highlights key information, enhances the model's ability to capture global information, and maintains efficient feature extraction. The module utilizes a four-layer 2-D convolutional architecture, with a SADB incorporated after the second and third convolutional layers.

For the critical SADB, the spectral dimension is initially downscaled using a 1×1 convolution. Since HSI

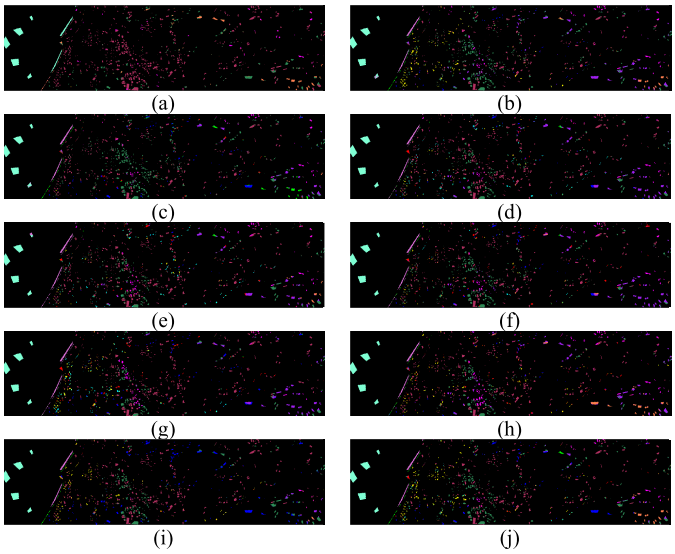


Fig. 8. Classification maps of different methods for Loukia. (a) DAAN. (b) DSAN. (c) TVT. (d) DDAN-JCA. (e) SDEnet. (f) S2ECNet. (g) TSTNet. (h) LLURNet. (i) S2AMNet. (j) ASAL.

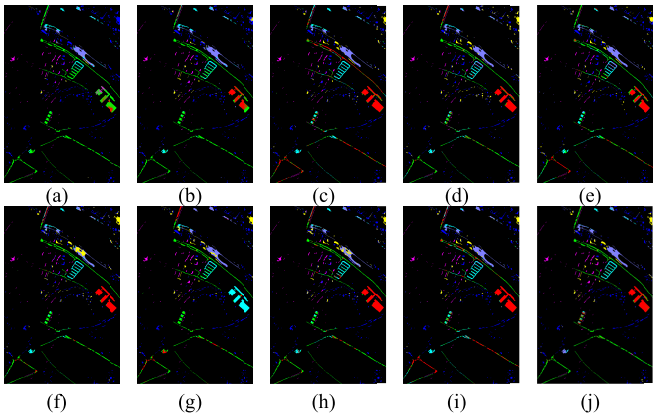


Fig. 9. Classification maps of different methods for PC. (a) DAAN. (b) DSAN. (c) TVT. (d) DDAN-JCA. (e) SDEnet. (f) S2ECNet. (g) TSTNet. (h) LLURNet. (i) S2AMNet. (j) ASAL.

data typically contain high-dimensional spectral channels, this dimensionality reduction effectively decreases the computational burden while preserving essential spectral information

$$\mathbf{X}_1 = \mathrm{Conv}_{1\times 1}(\mathbf{X}). \tag{7}$$

After spectral dimensionality reduction, depthwise separable convolution (DSC) [58] is introduced as the bottleneck layer, which contains two main steps: depthwise (DW) convolution and pointwise (PW) convolution. It can efficiently extract local features in the spatial dimension while avoiding the high computational cost of traditional convolution

$$\mathbf{X}_2^{\text{depthwise}} = \mathbf{X}_1 * \mathbf{K}_c \tag{8}$$

$$\mathbf{X}_2 = \mathbf{X}_2^{\text{depthwise}} * \mathbf{K}_p \tag{9}$$

where $*$ is the convolution operation, $\mathbf{K}_c$ is the convolution kernel applied to the cth input channel. In the DW convolution step, we convolve each input channel independently. $\mathbf{K}_p$ is a 1×1 convolution kernel in the shape of

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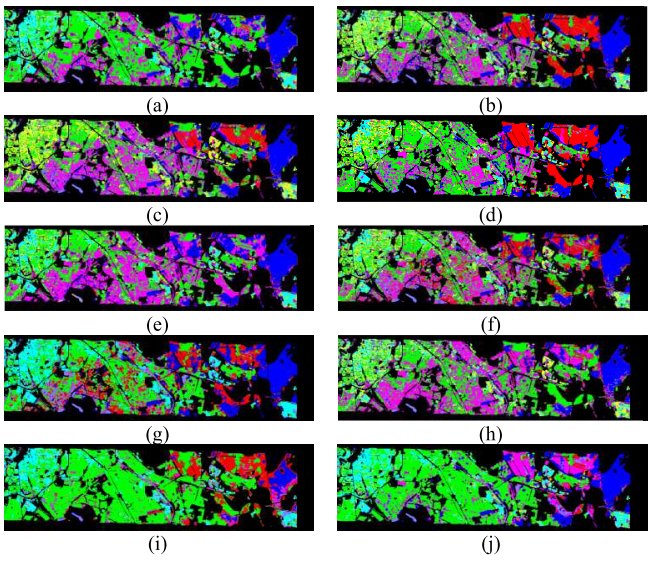


Fig. 10. Classification maps of different methods for Berlin. (a) DAAN. (b) DSAN. (c) TVT. (d) DDAN-JCA. (e) SDEnet. (f) S2ECNet. (g) TSTNet. (h) LLURNet. (i) S2AMNet. (j) ASAL.

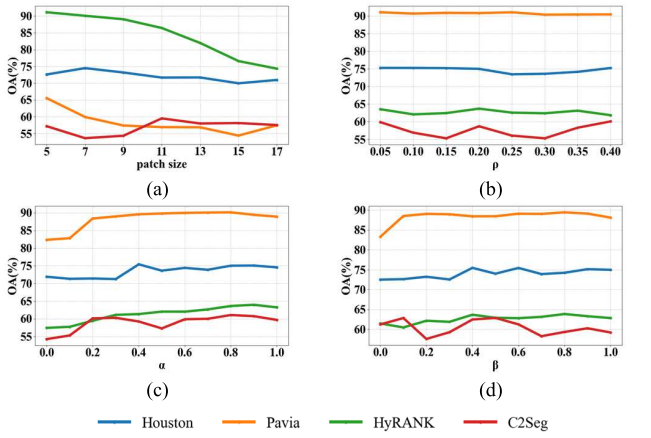


Fig. 11. Parameter analysis of the ASAL method. (a) Parameter patch size. (b) Parameter ρ. (c) Parameter α. (d) Parameter β.

$C_{\text{in}} \times C_{\text{out}}$. A new output channel C_{out} is generated across all C_{in} channels.

Next, batch normalization (BN) [59] is applied along the spectral dimension to balance the distribution of features, stabilizing the training process and addressing spectral dimension imbalances that can arise during training. Following normalization, we employ the sigmoid-weighted linear unit (SiLU) [60] activation function, which combines its inputs with the sigmoid function to achieve enhanced nonlinear properties. The SiLU activation function is particularly valued for its smoothness and interpretability

$$\mathbf{X}_3 = \text{SiLU}\left(\text{BatchNorm}\left(\mathbf{X}_2\right)\right) \tag{10}$$

where $\mathbf{X}_2$ is the output of the DSC.

HSIs contain intricate patterns and details. To improve the model's feature extraction capability and mitigate the risk of gradient vanishing during training, we introduce DC [61] after DSC, and employ a residual connection to retain the original feature information, further enhancing the model's learning capacity.

Compressed contextual features are obtained using DSC. The residual connection further aids the network in adapting to varying input data, enhancing its generalization to new data—especially TD data—and facilitating the learning of fine-grained features. In addition, residual connections support complex feature mapping, empowering the network to handle the intricate characteristics of HSI data more effectively. Subsequently, DC enhances the model's spatial adaptability, which is particularly crucial for cross-domain classification of HSI data, where spatial features can vary significantly across different scenes. DC dynamically adjusts the convolution kernel's sampling positions, enabling the model to flexibly capture spatial information in diverse environments and improve its adaptability to complex scenes

$$\mathbf{X}_4 = \mathbf{X}_3 + \mathbf{W}_r \cdot \mathbf{X}_3 \tag{11}$$

$$\mathbf{X}_{i,j,k} = \sum_{l=1}^{k_h \cdot k_w} \sum_{m=1}^{C_{\text{in}}} \mathbf{X}_4 \left(i + \Delta p_{i,l}, j + \Delta p_{j,l}, m\right) \cdot \mathbf{W}_{l,m,k} \tag{12}$$

where $\mathbf{W}_{\text{r}}$ is the residual convolution layer, which performs a linear transformation of the input features. $\Delta p_{i,l}$ and $\Delta p_{j,l}$ are the offsets of the convolution kernel at position (i, j). $W_{l,m,k}$ is the weight of the DC kernel.

Layer normalization (LN) [62] follows the DC and sigmoid activation function to further process the deformed features X_5, enhancing their stability across different spatial dimensions:

$$A = \text{Sigmod}\left(\text{LayerNorm}\left(X_5\right)\right). \tag{13}$$

The output of (13) represents the attention gate, which regulates the flow of information in the feature map. Each element in the attention gate has a value between 0 and 1, determining which regions in the feature map will be highlighted or suppressed. Finally, to incorporate the SDAB into the feature extractor, we perform a PW multiplication of the original input tensor with the attention tensor derived from the previous step

$$X_{\text{out}} = X_1 \odot A \tag{14}$$

where X_1 is the input feature map, A is the attention weight.

While selecting the feature extractor, we integrate the attention mechanism into the top layers of the CNN. To better adapt to CNN, a 3×3 convolution is used instead of the fully connected layer to compute offsets. A fully connected layer typically breaks the image's spatial structure by treating every pixel location of the input feature map as fully connected. In contrast, the convolutional layer uses the same kernel across the entire input image, maintaining the local spatial relationships between features. This approach ensures consistent feature extraction at different locations, which is crucial for handling spatially structured data.

C. EMC Loss

In classification tasks, category confusion arises when a model misclassifies samples from one category as belonging to another. This issue is particularly prominent in situations with significant category similarity or imbalanced data distributions.

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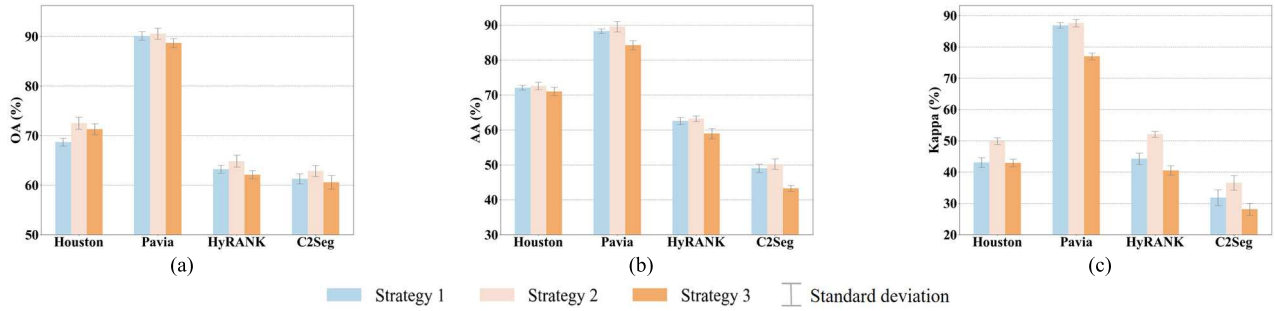


Fig. 12. Results for different numbers of SADB layers. (a) OA. (b) AA. (c) Kappa. In the legend, Strategy 1 refers to adding the attention mechanism once in the convolutional layer, Strategy 2 refers to adding it twice, and Strategy 3 refers to adding it three times. The graph shows the maximum value of accuracy for a fixed number of cases.

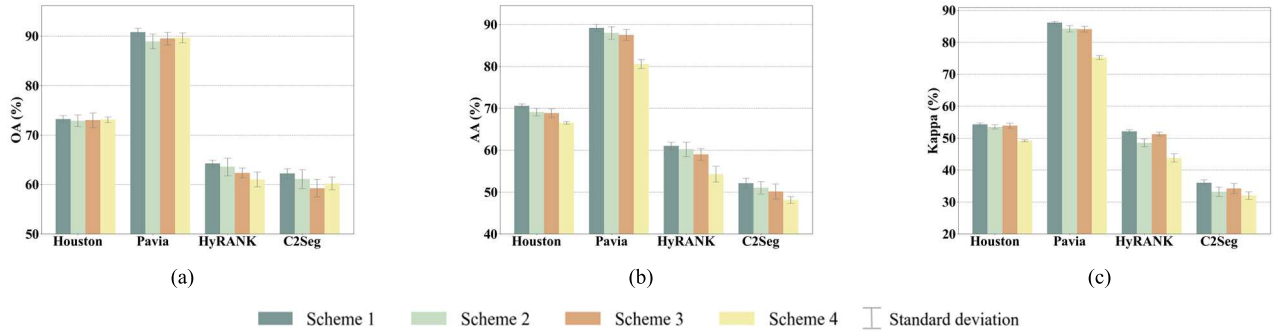


Fig. 13. Results for different SADB positions. (a) OA. (b) AA. (c) Kappa. In the legend, Scheme 1 refers to adding the attention mechanism after the convolution of the second and third layers, Scheme 2 refers to the first and second layers, Scheme 3 refers to the first and third layers, and Scheme 4 refers to the third and fourth layers.

In the context of HSI cross-domain classification, the challenge is further compounded by the absence of TD labels, making it impossible to directly calculate ground-truth category confusion for the TD. To address this limitation, we introduce a novel approach that employs inner product operations to quantify the similarity or confusion between categories. By calculating the instance-weighted inner product between the predicted class probability matrix and its transpose, this method provides an intuitive and effective means of capturing intercategory confusion. This representation naturally reflects the degree of misclassification, offering valuable insights into the classification process. Building on this foundation, we propose the EMC Loss, a robust loss function designed to mitigate category confusion by penalizing inaccurate predictions. The conceptual workflow of the EMC Loss framework is outlined in Fig. 2.

The classification accuracy of modern neural networks has improved significantly, but their confidence calibration has deteriorated. To ensure that the predictive information of a classifier is not biased toward certain classes, it is crucial to obtain well-calibrated confidence scores that accurately reflect the actual likelihood of correctness. Temperature scaling is a straightforward and effective method for achieving this. It adjusts the model's outputs by dividing the logits by a learned temperature parameter. This process softens the predicted probabilities, making overconfident predictions align more closely with the actual likelihood of correctness. Importantly, temperature scaling does not change the model's predictions but adjusts the confidence levels to make them more realistic. The scaling of the logits is achieved by introducing the temperature parameter T, with the adjusted softmax function [47] expressed as follows:

$$\hat{p}_i = \frac{\exp\left(\frac{Z_i}{T}\right)}{\sum_{j=1}^{C} \exp\left(\frac{Z_j}{T}\right)} \tag{15}$$

where $\mathbf{Z}_i$ is the classifier's output and C is the total number of categories.

The classifier is considered less effective when its predictions approach a uniform distribution, lacking distinct peaks (i.e., when no class is significantly more probable than others). Conversely, when predictions exhibit multiple peaks, it indicates that the classifier is uncertain between several ambiguous classes.

To address this, entropy weights are introduced to quantify the model's confidence level for different samples. Information entropy serves as a measure of uncertainty, reflecting the varying amounts of information provided by different categories of samples. This variability influences the relevance of each sample's contribution to the classification task. High entropy reflects greater uncertainty in the model's class assignments. Low entropy indicates higher confidence in the model's predictions. By incorporating entropy weights, the model can focus more effectively on learning from challenging and ambiguous samples. During the training process, samples with higher uncertainty (and thus higher entropy) are

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assigned greater importance, enabling the model to refine its handling of difficult cases. The target entropy is expressed as

$$H(p_i) = -\sum_{j=1}^{C} p_{ij} \log\left(p_{ij}\right) \tag{16}$$

where $H(p_i)$ denotes the information entropy of the predictive distribution p_{ij}, and p_{ij} is the predicted probability of the ith sample belonging to the jth class.

The entropy weight can be expressed as

$$W_{ii} = \frac{B\left(1+\exp\left(-H\left(\mathbf{p}_i\right)\right)\right)}{\sum_{j=1}^{B}\left(1+\exp\left(-H\left(\mathbf{p}_j\right)\right)\right)} \tag{17}$$

where W_{ii} denotes the entropy weight used for the ith sample and B denotes the batch size.

HSI data typically contain abundant spectral information, with features from different categories exhibiting distinct characteristics at varying scales. Analyzing category information across multiple scales allows the model to better understand the details and complexity of each category, leading to improved classification accuracy. In addition, multiscale analysis captures intercategory relationships, offering a broader perspective on correlations and variations between categories. This is particularly crucial for cross-domain classification tasks, where category distributions can vary significantly between the SD and TD. By incorporating multiscale analysis, the model can adapt more effectively to these differences, enhancing classification performance.

First, the probability distributions of the categories are weighted at each scale. Then, a covariance matrix is computed for each scale based on the weighted category distributions. Finally, these covariance matrices are merged to form the final multiscale covariance matrix. This integration balances information across scales, providing a more comprehensive statistical representation of the categories. This enhanced understanding supports improved categorization within the model

$$\sum_{s} = \frac{1}{B}\sum_{i=1}^{B} W_i\left(\hat{\mathbf{p}}_i - \bar{\hat{\mathbf{p}}}\right)\left(\hat{\mathbf{p}}_i - \bar{\hat{\mathbf{p}}}\right)^{\mathrm{T}} \tag{18}$$

$$\sum = \frac{1}{|S|}\sum_{s\in S}\sum_{\mathrm{s}} \tag{19}$$

where $\bar{\hat{p}}$ is the weighted mean of the sample predictions, $\sum_s$ is the calculated covariance matrix, $\sum$ is the combined covariance matrix, and S is the set of scales.

The calculated covariance matrix effectively captures the interclass relationships based on the above derivation. This approach combines entropy-weighted class probability and multiscale covariance analysis, focusing on cross-domain discriminative features. The EMC loss is defined as

$$\mathcal{L}_{\mathrm{EMC}} = \frac{\sum_{i=1}^{C}\sum_i - \mathrm{Tr}\left(\sum\right)}{C} \tag{20}$$

where $\sum_i$ is the ith diagonal element of the overall covariance matrix and $\mathrm{Tr}\left(\sum\right)$ is the trace of the overall covariance matrix, which represents the sum of all class variances.

D. Smooth Task Loss

The loss functions of DL models are typically highly complex in the parameter space, often containing numerous steep local minima [63]. Although these local minima can enable strong performance on the training set, their sharp gradient variations make the model insufficiently robust to small input perturbations. Consequently, this lack of robustness leads to poor generalization when the model is applied to the test set or TD. By smoothing the task loss, the model shifts its focus from merely minimizing loss at a sharp local optimum to finding a smoother minimum with reduced variation in loss over a broader range of parameters.

For the classification task in the SD, the classification loss can be expressed as $\mathcal{L}_{\mathrm{task}}(h_\theta(x), y)$. Traditional cross-entropy loss depends on one-hot label distributions, which can cause the model to overfit to the training samples, particularly when there are distributional differences between the SD and TD, potentially reducing the model's generalization ability. Label smoothing effectively mitigates the overfitting problem and improves the robustness of the model to samples in the TD by introducing a smoother target distribution. Label smoothing adjusts the true label distribution y to make it smoother, defined as

$$y_{\mathrm{smooth}} = (1-\delta)\, y + \frac{\delta}{C} \tag{21}$$

where δ is a label smoothing parameter that controls the smoothing intensity. δ is generally taken between 0.01 and 0.1, and the best result from [0.01,0.05,0.1] is used in the dataset.

The basic objective of domain adversarial training is to minimize the classification loss on the SD, while aligning the feature distributions of the SD and TD using an adversarial loss. The overall objective can be expressed as

$$\min_\theta \max_\Phi \mathrm{E}_{x\sim\mathcal{D}_S}\left[\mathcal{L}_{\mathrm{task}}\left(h_\theta(x), y\right)\right] + \mathcal{L}_{\mathrm{adv}}\left(h_\theta\left(x^S\right), h_\theta\left(x^{\mathrm{T}}\right), \Phi\right) \tag{22}$$

where h_θ represents the feature extractor, $\mathcal{L}_{\mathrm{task}}$ is the classification loss on the SD, and $\mathcal{L}_{\mathrm{adv}}$ is the adversarial loss designed to align the source and target feature distributions.

We propose the LSDA method to achieve a smoothed minimum task loss, enhancing generalization to the TD. Inspired by the SAM approach, LSDA identifies smooth minima by considering not only the task losses at the current model parameters but also those in the surrounding neighborhood during the optimization process. The gradient of the model's current parameters is first computed, followed by a small perturbation in the gradient direction, after which the loss is recalculated. The final optimization is performed using this perturbed loss, ensuring reduced loss in the local regions of the parameter space while avoiding sharp loss surfaces. For each model parameter θ, we define a local perturbation ϵ to guide the model toward a smooth minimum of the task loss. The modified objective function can be expressed as

$$\begin{aligned}\min_\theta \max_\Phi \max_{\|\epsilon\|\le\rho} \mathrm{E}_{x\sim\mathcal{D}_S}\left[\mathcal{L}_{\mathrm{task}}\left(h_{\theta+\epsilon}(x), y_{\mathrm{smooth}}\right)\right] \\ + \mathcal{L}_{\mathrm{adv}}\left(h_\theta\left(x^S\right), h_\theta\left(x^{\mathrm{T}}\right), \Phi\right)\end{aligned} \tag{23}$$

where ϵ is a perturbation bounded by ρ, controlling the extent of smoothness applied to the task loss. The adversarial

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TABLE I
LAND COVER CATEGORIES AND SAMPLE NUMBERS IN THE HOUSTON DATASET

Class		Numbers of samples	
No.	name	Houston13	Houston18
1	Grass healthy	345	1353
2	Grass stressed	365	4888
3	Trees	365	2766
4	Water	285	22
5	Residential buildings	319	5347
6	Non-residential buildings	408	32459
7	Road	443	6365
	Total	2530	53200

loss $\mathcal{L}_{\text{adv}}$ remains unchanged to preserve the effectiveness of domain alignment.

The TD risk $R_T(h_\theta)$ in LSDA can be upper bounded by

$$R_T(h_\theta) \le \max_{\|\epsilon\|\le\rho} \hat{R}_S\left(h_{\theta+\epsilon}\right) + D_\Phi\left(\mathcal{D}_S \| \mathcal{D}_T\right) + \gamma\left(\frac{\|\theta\|_2^2}{\rho^2}\right) + \lambda^* \tag{24}$$

where $\hat{R}_S(h_{\theta+\epsilon})$ is the empirical source risk with the perturbation, $D_\Phi(\mathcal{D}_S\|\mathcal{D}_T)$ is the domain discrepancy, and λ^* is the irreducible error. The additional regularization term $\gamma(\|\theta\|_2^2/\rho^2)$ ensures that the model parameters do not grow excessively large, further improving stability and performance.

LSDA is usually performed in two stages: decoupling the classification task and optimizing domain alignment. In the first stage, we optimize the SD classification task and learn the classifiers and basic feature extractors to ensure the model has sufficient classification capability. In the second stage, we add adversarial loss to align the source and TD feature distributions through gradient inversion or adversarial optimization, while fine-tuning the classifiers to fit the TD. The two-stage equations are as follows:

$$\mathcal{L}_1 = \alpha\mathcal{L}_{\text{task}} + (1-\alpha)\,L_{\text{EMC}} \tag{25}$$

$$\mathcal{L}_2 = \beta\mathcal{L}_{\text{task}} + (1-\beta)\,L_{\text{adv}} \tag{26}$$

where α and β are the weights of losses for the first stage and the second stage tasks, respectively.

IV. EXPERIMENTAL RESULTS AND ANALYSIS

A. Dataset Descriptions and Evaluation Indexes

Three publicly available HSI datasets, Houston, Pavia, and HyRANK are used to evaluate the proposed method's effectiveness.

1) Houston [64], [65]: The dataset is collected at the University of Houston campus in Houston, TX, USA, using different sensors, including Houston 2013 (SD) and Houston 2018 (TD). Houston 2013 contains 349 × 1905 pixels with 144 bands. Houston 2018 includes 209 × 955 pixels with 48 bands. On the Houston 2013 dataset, we select 209 × 955 overlapping regions and 48 spectral bands corresponding to Houston 2018. They all contain seven categories, and the details of the samples are shown in Table I. Fig. 3 shows their false-color images and ground-truth maps.

TABLE II
LAND COVER CATEGORIES AND SAMPLE NUMBERS IN THE HYRANK DATASET

Class		Numbers of samples	
No.	name	Dioni	Loukia
1	Dense Urban Fabric	1262	206
2	Mineral Extraction Sites	204	54
3	Non Irrigated Arable Land	614	426
4	Fruit Trees	150	79
5	Olive Groves	1768	1107
6	Coniferous Forest	361	422
7	Dense Sclerophyllous Vegetation	5035	2996
8	Sparce Sclerophyllous Vegetation	6374	2361
9	Sparsely Vegetated Areas	1754	399
10	Rocks and Sand	492	453
11	Water	1612	1393
12	Coastal Water	398	421
	Total	20024	10317

TABLE III
LAND COVER CATEGORIES AND SAMPLE NUMBERS IN THE PAVIA DATASET

Class		Numbers of samples	
No.	name	UP	PC
1	Tree	3064	7598
2	Asphalt	6631	9248
3	Brick	3682	2685
4	Bitumen	1330	7287
5	Shadow	947	2863
6	Meadow	18649	3090
7	Bare soil	5029	6584
	Total	39332	39335

2) HyRANK [66]: The dataset consists of data from two regions, Dioni (SD) and Loukia (TD). They are acquired by the Hyperion sensor (EO-1, USGS) with 176 spectral bands. The sizes of Dioni and Loukia are 250 × 1376 pixels and 246 × 945 pixels, respectively. There are 12 common categories listed in Table II. The two scenes are represented in pseudo-color, and the ground surface is represented in pseudo-color. Pseudo-color representations and ground-truth maps of the two scenes are shown in Table II and Fig. 4.

3) Pavia [67]: This dataset consists of the University of Pavia (UP, SD) and the Pavia Center (PC, TD), both acquired by ROSIS sensors located in Pavia, northern Italy. The UP has 610 × 610 pixels with 103 bands, and the PC has 1096 × 1096 pixels with 102 bands. After cropping according to the experimental needs, UP contains 610 × 315 pixels, and PC contains 1096 × 715 pixels. The last band was removed for UP to ensure the same number of spectral bands as PC. They both have seven identical categories, and the details of the samples are shown in Table III. Fig. 5 shows their false-color images and ground-truth maps.

4) C2Seg-AB [68]: The dataset consists of data from two regions, Augsburg (SD) and Berlin (TD). They are captured by EnMAP, with Berlin covering 2465 × 811 pixels and Augsburg covering 886 × 1360 pixels at 10-m GSD. The ground truth contains 13 identical land cover categories for

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TABLE IV
LAND COVER CATEGORIES AND SAMPLE NUMBERS IN THE C2SEG-AB DATASET

Class		Numbers of samples	
No.	name	Augsburg	Berlin
1	Forest	13627	54954
2	Residential Area	42819	268642
3	Industrial Area	7045	19566
4	Low Plants	34938	59282
5	Allotment	2254	13305
6	Commercial Area	2576	24824
7	Water	2131	6672
Total		105390	447245

both cities, including forests, water, and street networks, derived from OpenStreetMap with manual refinement. Pseudo-color representations and ground-truth maps of the two scenes are shown in Table IV and Fig. 6.

B. Experimental Setup

Several state-of-the-art transfer learning algorithms are used for comparison. These include the DAAN [34], deep sub-domain adaptive network (DSAN) [69], transferable vision transformer (TVT) [70], deep dynamic adaptation network with joint CORAL (DDAN-JCA) [71], single-SD expansion network (SDEnet) [72], the spatial–spectral enhancement and causal constraint network (S2ENet) [73], a topological structure and semantic information transfer network (TSTNet) [74], locally linear unbiased randomization network (LLURNet) [75], and spectral–spatial adversarial multidomain synthetic network(S2AMNet) [76].

All experiments are conducted on the same hardware platform, featuring an NVIDIA GeForce RTX 4090 GPU and an Intel i5-11400H CPU. To ensure fairness across experiments, 180 samples are used for each category in the Houston, HyRANK, and Pavia datasets, and 1000 samples are used for each category in the C2Seg-AB dataset. Adaptive smooth adversarial learning (ASAL) is trained for 100 epochs with a batch size of 64 and a learning rate of 0.001.

1) *DAAN:* The model is trained for 100 epochs with a batch size of 64 and a learning rate of 0.001. The optimizer used is SGD with a momentum of 0.9 and a weight decay of 0.0005. For all datasets, the input patch size is 5×5.
2) *DSAN:* The model is trained for 100 epochs with a batch size of 64 and a learning rate of 0.001. The optimizer is SGD with moments of 0.9 and weight decay of 0.0001. For all datasets, the input patch size is 5×5.
3) *TVT:* The model is trained for 100 epochs with a batch size of 64. The optimizer used is SGD with a momentum of 0.9 and a weight decay of 0.0001. The learning rate is initially set to 0 and increases linearly to 0.001 after 20 training steps. A cosine decay strategy is used to reduce it. For all datasets, the input patch size is 5×5.
4) *DDAN-JCA:* The model is trained for 100 epochs with a batch size of 64. The parameter α is set to 0.5 for Pavia and 1 for Houston and HyRANK. For all datasets, the input patch size is 5×5.
5) *SDEnet:* The model is trained for 400 epochs with a batch size of 64 and a block size of 13×13. For all datasets, λ_1, λ_2, and the embedding feature dimension d_{se} are set to 1, 0.1, and 64, respectively.
6) *S2ECNet:* The model is trained for 400 epochs with a batch size of 64 and a learning rate of 0.001. The Adam optimizer is used, and the regularization parameter λ is set to 1. The patch size for all datasets is 13×13.
7) *TSTNet:* The model is trained for 500 epochs with a batch size of 64 and a learning rate of 0.001. The optimizer used is SGD with a momentum of 0.9. The L_2 regularization is set as $1e^{-4}$. The patch size for all datasets is 13×13.
8) *LLURNet:* The model is trained for 100 epochs with a batch size of 64 and a learning rate of 0.001. The optimizer is SGD with moments of 0.9 and weight decay of 0.0001. The embedded feature dimension $\dim_1$ and $\dim_2$ are set to 8 and 16, respectively. The patch size for all datasets is 13×13.
9) *S2AMNet:* The model is trained for 100 epochs with a batch size of 64 and a learning rate of 0.001. The temperature parameter in the InfoNCE loss function is set to 0.07. The patch size for all datasets is 13×13.

Overall accuracy (OA), average category accuracy (AA), and the kappa coefficient are used to quantitatively assess the classification performance. In order to obtain unbiased results, all experiments are repeated five times, and the average of the evaluated metrics is used as a comprehensive analysis of the classification performance.

C. Classification Results

Tables V–VIII show OA, AA, and the kappa coefficient for different comparison methods across four datasets, along with the classification accuracy for each class (highest accuracy in bold and second highest underlined). Figs. 7–10 show the visualization results of the four datasets. All experiments are repeated five times, and the average of the evaluated metrics is used as a comprehensive analysis of the classification results.

1) Overall Performance Comparison: Our method, ASAL, significantly enhances classification performance across the Houston, HyRANK, Pavia, and C2Seg-AB datasets, as detailed in Tables V–VIII. Specifically, ASAL achieves 76.03% OA in the Houston dataset, surpassing the performance of existing methods. In the HyRANK dataset, ASAL reaches 65.54% OA, while in the Pavia dataset, it attains 91.34% OA, showcasing its robust adaptability and generalization across varied domains. In the C2Seg-AB dataset, our method achieves an OA of 63.07%, outperforming the second-best method TSTNet (60.72%) by 2.35 percentage points. These results highlight ASAL's superior performance over other approaches, including DAAN, DSAN, and TVT, across multiple performance metrics—namely, OA, AA, and the Kappa coefficient. The consistent improvements across all datasets validate ASAL's effectiveness in achieving higher classification accuracy, particularly in the presence of domain shifts.

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TABLE V
EXPERIMENTAL RESULTS OF THE HOUSTON DATASET

Class	DAAN	DSAN	TVT	DDAN-JCA	SDEnet	S2ECNet	TSTNet	LLURNet	S2AMNet	ASAL
1	49.27 ± 39.41	54.47 ± 27.10	**82.8 ± 8.65**	63.09 ± 14.35	35.76 ± 12.52	9.58 ± 5.49	74.41 ± 24.17	27.70 ± 11.70	15.20 ± 28.83	79.44 ± 9.69
2	71.51 ± 16.59	**84.09 ± 6.79**	50.05 ± 12.21	49.72 ± 8.30	71.63 ± 5.54	59.10 ± 19.52	54.72 ± 22.86	72.07 ± 5.40	47.59 ± 32.09	82.92 ± 5.09
3	40.74 ± 25.21	47.83 ± 20.50	59.45 ± 7.26	43.03 ± 14.32	49.80 ± 8.57	33.90 ± 22.90	59.38 ± 10.51	**67.73 ± 4.54**	8.34 ± 9.20	61.00 ± 5.11
4	75.45 ± 22.75	86.36 ± 13.79	16.67 ± 6.09	40.00 ± 7.93	97.27 ± 5.45	98.29 ± 1.82	99.09 ± 2.73	**100.00 ± 0.00**	32.27 ± 22.49	96.97 ± 4.29
5	46.18 ± 36.34	70.63 ± 8.81	90.97 ± 4.24	89.78 ± 8.14	65.21 ± 12.75	50.01 ± 4.11	82.89 ± 14.45	66.77 ± 8.87	23.02 ± 14.00	**91.57 ± 4.22**
6	71.43 ± 17.52	61.13 ± 5.49	55.04 ± 8.32	70.58 ± 2.77	51.18 ± 5.32	66.61 ± 4.85	72.46 ± 10.58	58.40 ± 7.34	69.05 ± 27.06	**73.18 ± 2.87**
7	23.70 ± 26.73	62.38 ± 10.87	44.91 ± 13.27	31.07 ± 19.08	54.65 ± 14.12	55.54 ± 16.31	47.70 ± 23.50	70.83 ± 5.00	32.91 ± 26.20	**77.78 ± 1.74**
OA	62.27 ± 7.77	63.49 ± 4.32	58.1 ± 4.06	64.23 ± 5.64	54.43 ± 4.56	59.79 ± 2.79	70.44 ± 6.38	61.71 ± 4.41	53.59 ± 18.29	**76.03 ± 1.99**
AA	54.32 ± 10.41	66.70 ± 5.97	57.41 ± 6.42	55.32 ± 3.88	60.79 ± 4.60	53.40 ± 6.48	70.24 ± 7.31	66.21 ± 1.81	32.63 ± 10.23	**71.39 ± 1.49**
Kappa×100	37.94 ± 11.88	47.17 ± 5.44	42.09 ± 5.81	44.80 ± 3.72	37.23 ± 4.99	38.78 ± 5.58	53.06 ± 10.09	45.71 ± 4.22	35.86 ± 13.01	**59.09 ± 2.43**

TABLE VI
EXPERIMENTAL RESULTS OF THE HYRANK DATASET

Class	DAAN	DSAN	TVT	DDAN-JCA	SDEnet	S2ECNet	TSTNet	LLURNet	S2AMNet	ASAL
1	19.90± 24.70	33.76 ± 22.24	10.03± 6.27	13.79± 5.26	10.93 ± 8.73	34.05 ± 4.75	**43.69 ± 22.73**	43.45 ± 5.66	18.54 ± 9.26	43.36 ± 8.87
2	40.00± 46.05	61.85 ± 28.00	52.47± 25.68	50.00± 27.28	66.67 ± 32.61	65.56 ± 25.14	38.89 ± 47.66	63.70 ± 39.36	37.59 ± 35.38	**98.14 ± 0.15**
3	36.29± 44.21	51.64 ± 15.32	13.93± 6.38	47.18± 12.25	16.01 ± 14.92	28.83 ± 10.17	47.93 ± 16.98	37.21 ± 12.20	**67.61 ± 15.14**	59.78 ± 11.02
4	9.87± 18.51	64.81 ± 29.00	77.22± 1.29	**81.01± 17.42**	13.92 ± 10.83	23.04 ± 11.72	75.70 ± 12.05	34.30 ± 21.47	27.59 ± 16.02	77.63 ± 2.15
5	**33.17± 20.63**	10.26 ± 11.81	13.82± 6.29	15.01± 12.94	5.84 ± 3.66	6.81 ± 4.52	9.65 ± 12.50	3.20 ± 4.67	0.01 ± 0.03	18.60 ± 4.59
6	23.18± 26.43	46.49 ± 16.51	35.15± 18.49	63.08± 12.00	64.12 ± 15.14	53.08 ± 9.04	25.92 ± 13.08	55.02 ± 9.84	55.24 ± 16.95	**67.14 ±** 15.86
7	46.78± 40.02	66.15 ± 7.35	49.91± 5.36	51.66± 2.23	51.04 ± 27.89	71.29 ± 15.01	**71.68 ± 7.50**	64.07 ± 7.45	71.35 ± 12.31	50.04 ± 7.98
8	38.96 ± 26.13	**65.01 ± 3.84**	21.25± 18.15	35.52± 13.80	32.41 ± 22.04	46.38 ± 5.87	59.88 ± 11.29	48.59 ± 10.31	49.76 ± 11.51	42.05 ± 6.29
9	15.49 ± 27.27	**53.28 ± 27.23**	26.9± 18.49	46.22± 23.49	44.06 ± 33.07	51.33 ± 20.87	50.03 ± 21.62	36.12 ± 13.09	23.61 ± 12.43	14.77 ± 4.83
10	41.59 ± 34.67	38.41 ± 23.41	73.07± 1.17	43.4± 26.97	24.12 ± 29.65	6.90 ± 12.83	23.58 ± 24.70	24.66 ± 24.97	29.40 ± 26.17	**73.43 ± 0.21**
11	98.51 ± 2.95	96.55 ± 4.66	99.21± 0.65	98.75± 1.53	99.30 ± 1.41	**100.00 ± 0.00**	99.77 ± 0.46	**100.00 ± 0.00**	99.79 ± 0.28	**100.00 ± 0.00**
12	61.81 ± 45.77	95.67 ± 5.75	**100.00± 0.00**	99.57± 0.43	81.14 ± 37.71	**100.00 ± 0.00**	**100.00 ± 0.00**	**100.00 ± 0.00**	99.64 ± 1.07	**100.00 ± 0.00**
OA	47.44 ± 11.07	61.40 ± 3.58	45.64± 4.26	52.64± 3.44	46.30 ± 16.14	56.47 ± 2.74	60.77 ± 2.03	55.38 ± 3.94	57.68 ± 4.68	**65.54 ± 0.77**
AA	38.80 ± 7.90	56.99 ± 5.29	47.75± 3.65	53.77± 4.08	42.46 ± 13.22	48.94 ± 2.44	53.89 ± 4.76	50.86 ± 3.07	48.35 ± 2.72	**61.58 ± 1.18**
Kappa×100	37.72 ± 9.14	52.76 ± 4.59	37.63± 5.29	44.19± 3.59	38.00 ± 15.31	47.56 ± 2.59	52.54 ± 1.84	46.82 ± 4.10	48.95 ± 4.88	**53.78 ± 4.70**

2) Improvement of Feature Extraction Strategy: While DAAN struggles with extracting complex spectral and spatial features in HSIs, ASAL leverages advanced feature extraction strategies, resulting in superior classification performance. DSAN, although effective in DA through subdomain alignment, faces challenges when handling the high-dimensional and intricate nature of hyperspectral data. In the HyRANK dataset, ASAL outperforms DSAN by 12.82% OA and 20.65% OA in the classification of fruit trees (category 4) and coniferous forest (category 6), respectively. TVT, which employs a transformer-based attention mechanism, demonstrates limited success in capturing subtle spectral details due to the complexity of its structure. In the Pavia dataset, for the brick (category 3) and bitumen (category 4), the TVT method can

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TABLE VII
EXPERIMENTAL RESULTS OF THE PAVIA DATASET

Class	DAAN	DSAN	TVT	DDAN-JCA	SDEnet	S2ECNet	TSTNet	LLURNet	S2AMNet	ASAL
1	99.23 ± 0.33	92.03 ± 2.48	**99.26 ± 0.34**	96.08 ± 1.39	80.88 ± 5.53	84.58 ± 4.28	87.77 ± 8.06	80.06 ± 14.87	71.30 ± 23.36	94.22 ± 3.84
2	**94.32 ± 3.01**	87.77 ± 5.89	88.39 ± 2.86	92.24 ± 0.76	57.98 ± 9.91	64.51 ± 17.50	92.55 ± 3.82	71.11 ± 15.99	67.40 ± 20.00	87.61 ± 2.27
3	55.57 ± 29.73	85.88 ± 8.01	29.51 ± 24.67	75.23 ± 3.07	76.73 ± 12.93	86.64 ± 8.26	75.86 ± 6.94	71.51 ± 12.85	76.21 ± 18.22	**96.42 ± 4.30**
4	29.09 ± 31.74	71.65 ± 7.43	69.29 ± 7.06	45.96 ± 18.59	79.56 ± 6.24	83.67 ± 1.40	0.97 ± 0.34	**85.58 ± 2.49**	85.07 ± 2.20	85.54 ± 0.19
5	85.69 ± 5.19	91.81 ± 3.04	76.54 ± 2.05	98.68 ± 0.86	85.66 ± 9.99	80.01 ± 8.01	97.23 ± 1.37	88.70 ± 3.23	82.67 ± 4.67	**99.24 ± 0.08**
6	18.22 ± 8.05	56.93 ± 16.74	14.13 ± 4.53	70.20 ± 12.12	78.10 ± 11.96	72.24 ± 4.30	77.18 ± 9.70	76.10 ± 7.32	73.16 ± 16.84	**85.09 ± 2.87**
7	68.71 ± 19.87	76.08 ± 5.91	67.48 ± 9.21	81.78 ± 2.46	82.03 ± 2.98	78.89 ± 11.56	**86.65 ± 5.41**	76.59 ± 10.01	63.85 ± 28.16	80.33 ± 1.09
OA	69.66 ± 7.29	81.40 ± 1.35	73.66 ± 1.42	80.74 ± 4.84	75.63 ± 2.02	77.59 ± 3.02	71.68 ± 1.45	78.13 ± 7.30	73.00 ± 4.67	**91.34 ± 5.59**
AA	64.40 ± 7.95	80.31 ± 2.86	64.67 ± 3.10	80.56 ± 4.39	77.56 ± 4.18	78.65 ± 1.96	74.03 ± 1.98	78.52 ± 5.58	74.24 ± 4.87	**89.78 ± 0.97**
Kappa×100	62.94 ± 8.91	77.58 ± 1.73	66.40 ± 1.93	76.70 ± 5.89	70.98 ± 2.50	73.27 ± 3.51	66.64 ± 1.69	73.98 ± 8.41	68.04 ± 5.32	**86.27 ± 1.50**

TABLE VIII
EXPERIMENTAL RESULTS OF THE C2SEG-AB DATASET

Class	DAAN	DSAN	TVT	DDAN-JCA	SDEnet	S2ECNet	TSTNet	LLURNet	S2AMNet	ASAL
1	76.32 ± 2.68	81.88 ± 2.92	77.75 ± 9.50	**83.46 ± 5.30**	75.28 ± 2.12	73.78 ± 6.19	82.13 ± 4.92	69.16 ± 5.85	42.04 ± 8.87	80.73 ± 2.04
2	48.35 ± 12.68	38.54 ± 6.16	21.74 ± 8.83	21.06 ± 13.53	37.25 ± 4.79	40.59 ± 6.55	61.51 ± 10.58	36.78 ± 5.94	**75.84 ± 4.75**	18.78 ± 6.87
3	**53.27 ± 9.24**	30.78 ± 7.47	20.46 ± 21.88	41.35 ± 12.96	45.97 ± 6.01	39.79 ± 12.13	46.82 ± 23.97	32.83 ± 6.36	41.16 ± 7.14	29.70 ± 18.77
4	5.81 ± 1.87	53.98 ± 16.49	53.60 ± 15.13	47.01 ± 31.11	1.93 ± 1.21	8.79 ± 4.48	40.36 ± 19.24	24.11 ± 14.61	54.96 ± 18.79	**62.88 ± 17.73**
5	59.82 ± 11.15	80.17 ± 5.45	58.56 ± 19.61	44.23 ± 30.08	**87.01 ± 3.65**	83.73 ± 1.11	41.52 ± 30.42	71.18 ± 8.56	21.62 ± 16.96	34.85 ± 28.48
6	16.74 ± 2.72	37.94 ± 7.27	**42.78 ± 20.65**	32.01 ± 16.67	21.44 ± 5.95	25.70 ± 9.54	18.07 ± 19.69	30.92 ± 4.93	7.38 ± 3.44	38.58 ± 24.27
7	41.85 ± 6.69	73.77 ± 4.38	84.37 ± 1.52	90.12 ± 6.21	46.45 ± 22.78	50.78 ± 9.75	70.23 ± 15.79	74.70 ± 7.66	70.86 ± 4.09	**94.40 ± 3.29**
OA	44.86 ± 7.11	47.30 ± 3.10	46.87 ± 7.25	55.34 ± 5.06	38.36 ± 2.27	41.02 ± 3.02	60.72 ± 3.46	40.17 ± 3.89	56.25 ± 3.35	**63.07 ± 4.35**
AA	43.17 ± 1.66	51.72 ± 3.51	44.18 ± 5.69	51.32 ± 2.76	45.05 ± 3.42	46.16 ± 1.52	51.52 ± 5.46	48.53 ± 2.03	46.98 ± 4.24	**52.85 ± 4.71**
Kappa×100	25.99 ± 4.06	32.74 ± 3.30	28.29 ± 6.54	33.63 ± 5.13	22.50 ± 1.17	24.21 ± 1.63	31.51 ± 3.57	28.93 ± 2.91	35.06 ± 6.30	**36.58 ± 4.39**

only achieve 39.51% OA and 69.29% OA, while our method can achieve 96.42% OA and 85.54% OA. This discrepancy highlights the ability of ASAL to handle fine-grained spectral analysis, particularly in distinguishing building surface materials, which require careful attention to spectral differences. The key advantage of ASAL lies in its integration of an attention mechanism within a CNN-based architecture, which enables more efficient and precise spectral–spatial feature extraction. This approach allows ASAL to better manage cross-domain variations, leading to improved classification accuracy across different datasets and categories.

3) Category-Specific Improvements and Noise Reduction: ASAL demonstrates a marked improvement in category-specific classification accuracy and effectively mitigates category confusion, particularly in challenging categories. For instance, in the Houston dataset, ASAL achieves 91.57% OA for residential buildings (category 5), 73.18% OA for non-residential buildings (category 6), and 77.78% OA for roads (category 7). In the C2Seg-AB dataset, our method achieves higher accuracy on multiple categories, reaching 62.88% for the low plants class (category 4), which is 7.92 percentage points higher than the second-highest method (54.96%). These improvements are attributed to ASAL's category confusion minimization strategy, which reduces overconfident predictions and enhances the model's adaptability to difficult cases. The effectiveness of this strategy is further validated by classification result plots, which reveal that ASAL produces significantly less noise and category overlap. For example, in the TVT method, noise is more pronounced, and the DDAN-JCA and SDEnet models demonstrate a higher tendency to

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misclassify roads as nonresidential buildings. In contrast, ASAL's plots exhibit clearer boundaries between categories, reflecting the efficacy of its temperature scaling and entropy-based calibration techniques in reducing misclassifications and improving overall classification performance.

4) Importance of Network Optimization and Task Loss Design: Methods such as SDEnet leverage densely connected structures to enhance spectral and spatial feature extraction. However, these methods often introduce redundant information in hyperspectral data, which can lead to performance degradation. Similarly, TSTnet, reliant on dynamically constructed subgraphs from CNN features, remains sensitive to noise and ambiguity, particularly in mixed land cover regions. ASAL uses double smoothing to regularize the model in both feature space and label space, effectively addressing these challenges. In the HyRANK dataset, our method achieved 65.54% OA, which is about 4.1 percentage points ahead of the second-highest DSAN method (61.40%). By reducing the distribution discrepancy between source and TDs, ASAL not only enhances stability during training but also improves generalization to unseen target data. These results demonstrate ASAL's robust capability in addressing the real-world challenges of hyperspectral cross-domain classification.

Our method performs well on the Houston, HyRANK, Pavia, and C2Seg-AB datasets, significantly outperforming existing methods. This is attributed to our innovative cross-scene HSI classification framework, attentional convolution module, and category confusion minimization strategy, which improve the model's stability, generalization ability, and classification accuracy.

D. Ablation Study

This section examines the impact of each proposed structure on the classification performance across three representative datasets. The experimental results are presented in Table IX (highest accuracy in bold, second highest underlined).

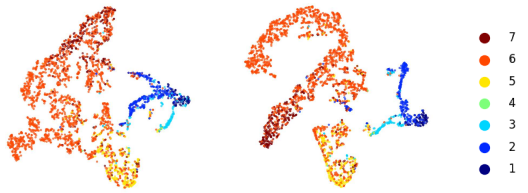
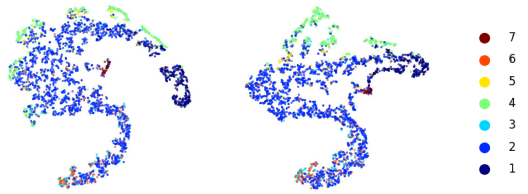
SADB indicates whether the attention mechanism is used; similarly, LSDA denotes whether the smoothed domain adversarial process is utilized, with a standard single-stage feedforward process used in its absence; and EMC indicates whether the EMC loss is used.

According to the experimental results, the attention mechanism significantly improves the classification performance. For example, on the HyRANK dataset, the application of the attention mechanism improves the OA from 62.15% to 65.54%, which is an improvement of 3.39%. Similarly, in the Houston and Pavia datasets, the addition of the attentional mechanism improved the OA by 1.88% and 1.52%, respectively. These results demonstrate the crucial role of the attention mechanism in capturing both spatial and spectral features of HSIs. In addition, LSDA significantly improves the classification results. Compared with the results without LSDA, the OA of the HyRANK dataset improves from 60.43% to 62.15%, which enhances the model's generalization capability over the TD by reducing the distributional difference between the SD and TD. Similarly, EMC significantly improves the classification performance. In the Pavia dataset, OA improves from 89.82% to 91.34% after applying EMC loss.

TABLE IX
ABLATION EXPERIMENTAL RESULTS

Dataset	SADB	LSDA	EMC	OA	AA	Kappa
Houston	×	×	×	68.65	67.15	52.13
	√	×	×	73.01	67.50	56.40
	×	√	×	72.62	69.23	<u>56.91</u>
	×	×	√	70.70	68.19	53.96
	×	√	√	74.15	<u>71.23</u>	54.40
	√	√	×	<u>75.78</u>	70.20	56.52
	√	√	√	**76.03**	**71.39**	**59.09**
Hyrank	×	×	×	59.48	52.79	41.04
	√	×	×	61.24	53.52	44.74
	×	√	×	60.71	53.37	44.94
	×	×	√	60.43	57.85	40.31
	×	√	√	62.15	<u>59.20</u>	46.09
	√	√	×	<u>63.91</u>	55.84	<u>50.03</u>
	√	√	√	**65.54**	**61.58**	**50.78**
Pavia	×	×	×	87.48	84.61	80.68
	√	×	×	88.12	85.34	82.57
	×	√	×	87.63	84.92	81.97
	×	×	√	88.64	87.60	83.98
	×	√	√	<u>89.82</u>	<u>87.74</u>	84.43
	√	√	×	89.31	87.64	<u>84.46</u>
	√	√	√	**91.34**	**89.78**	**86.27**
C2Seg -AB	×	×	×	54.28	38.58	25.58
	√	×	×	58.39	46.88	28.26
	×	√	×	57.38	44.13	29.43
	×	×	√	58.22	46.37	29.39
	×	√	√	60.27	<u>51.13</u>	<u>34.91</u>
	√	√	×	<u>61.38</u>	50.24	32.16
	√	√	√	**63.07**	**52.85**	**36.58**

TABLE X
VISUALIZATION OF THE DISTRIBUTION OF FEATURES BEFORE AND AFTER APPLYING EMC LOSS

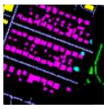
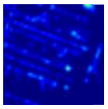
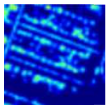
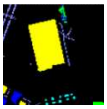
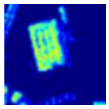
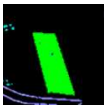
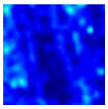
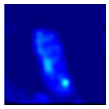
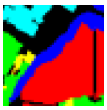
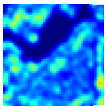
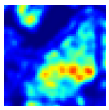
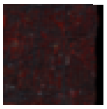
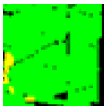
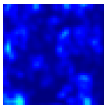
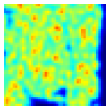
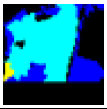
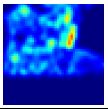
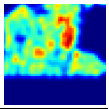
Dataset	No EMC loss	Add EMC loss
Houston		
C2Seg-AB		

Retaining all three modules yields the optimal classification results, underscoring the effectiveness of our algorithms in enhancing classification accuracy.

To illustrate the impact of the EMC loss on mitigating category confusion, we provide visualizations of the feature distributions before and after applying EMC. Different colors represent different categories. The results of the Houston dataset and the C2Seg-AB dataset are shown in Table X. The Houston dataset is small in scale and has a relatively balanced category distribution, while the C2Seg-AB dataset is large in scale and has an imbalanced category distribution. The two

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TABLE XI
VISUALIZATION OF ATTENTION MAPS BEFORE AND AFTER APPLYING ATTENTION

Dataset	Class name	false-color	ground-truth	before attention	after attention
Houston	Residential buildings				
	Non-residential buildings				
	Grass stressed				
	Low Plants				
C2Seg-AB	Residential Area				
	Industrial Area				

datasets are complementary in scale and category distribution and are representative.

For the Houston dataset, it can be clearly observed that after adding EMC loss, the distinction between category 6 (bright red) and category 7 (dark red) is significantly enhanced. Originally, these two categories had a large overlap in the feature space. After adding EMC loss, they gradually gathered to their respective centers, the boundaries between categories became clearer, and the clustering structure became more compact. This shows that EMC loss helps to enhance the model's ability to distinguish similar categories and optimize the discriminability and compactness of feature representation.

To explain the behavior of the SADB attention mechanism, we include attention heatmaps that highlight how the model focuses on semantically relevant regions during classification. In the heat map, cool areas represent areas where the model pays less attention, while warm areas correspond to areas where the model pays more attention. The results of the Houston dataset and the C2Seg-AB dataset are shown in Table XI.

For the Houston dataset, it can be clearly observed that after the introduction of attention, the model shows more focused attention in category-related areas. For the residential buildings category, the model clearly focused on the edge structures and dense areas of the residential area, showing a stronger ability to perceive spatial structures. For the grass stressed category, after adding attention, the model can more accurately focus on the grass stressed areas of large blocks, suppressing the response of the surrounding interference areas.

For the C2Seg-AB dataset, in the low plants category, the model can effectively focus on a large area of low plants and significantly suppress the attention response to the surrounding interference areas. In the residential area and commercial area categories, the model significantly increases the attention weight of the main target area, showing good responsiveness and strong focusing characteristics.

In summary, the introduced attention mechanism shows excellent spatial perception ability in different datasets and land object types. It can adaptively adjust the focus area according to the target category, making the attention response more in line with the real semantic area.

E. Parameter Sensitivity Analysis

The proposed model has four parameters: patch size, ρ, α, and β. Patch size refers to the size of the input HSI data, ρ from (23) defines the maximum magnitude of the SAM perturbation, α from (25) refers to the percentage of classifier loss in the first stage, and β from (26) refers to the percentage of classifier loss in the second stage.

For the patch size, the OA of the Houston dataset shows a slight decrease as the patch size increases, but the change is not particularly significant. This is due to the balanced spectral information and relatively uniform distribution of feature classes in the Houston dataset, allowing the model to fully leverage the spectral features across different patch sizes without overly depending on spatial information. In contrast, for the Pavia dataset, the OA decreases significantly with an increase in patch size, especially when the patch size reaches 15 and 17, where accuracy drops sharply to around 80%. This is because the Pavia dataset, which represents an urban scene with complex feature classes and less background redundancy, introduces more background information as the patch size increases. Since these background regions lack distinguishing features, they may introduce noise and negatively impact classification accuracy. For the C2Seg-AB dataset, OA first improves and then drops as the patch size increases. A moderately larger patch offers useful contextual cues to distinguish similar classes, but overly large patches may include mixed land cover types, leading to label ambiguity, increased noise, and degraded performance, especially for fragmented classes. The optimal patch sizes for Houston, HyRANK, Pavia, and C2Seg-AB are 7, 5, 5, and 11, respectively.

As shown in Fig. 11(b), the overall performance of the three datasets changes smoothly with little fluctuation as ρ increases, indicating that the model is not highly sensitive to this parameter and exhibits good robustness. There are no significant fluctuations in performance within the specified perturbation range. The optimal values for Houston, HyRANK, Pavia, and C2Seg-AB are 0.20, 0.20, 0.05, and 0.20, respectively.

For α and β, the weight proportions of each component of the loss function are adjusted to achieve optimal classification results. In our experiments, we test several combinations of these parameters. Initially, both parameters are set to 0.5. Then, α is adjusted while keeping β fixed, followed by further fine-tuning of β using the optimal α to ensure the best model performance. Ultimately, the optimal combinations of α and β are (0.4, 0.6), (0.9, 0.4), (0.8, 0.8), and (0.9, 0.5)

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for the Houston, HyRANK, Pavia, and C2Seg-AB datasets, respectively.

F. Attention Mechanisms Setup

To optimize the performance of the feature extractor module, we introduce an attention mechanism between the four convolutional layers. We conduct two sets of experiments: one to test the number of layers of attention and the other to test the position of attention.

The experimental results shown in Fig. 12 reveal that using two layers of the attention mechanism yields the best performance. For instance, on the Houston dataset, the OA, AA, and Kappa scores are 73.56%, 70.74%, and 44.49%, respectively. These results outperform other configurations, with the Kappa value being notably higher by 6%, indicating strong classification performance.

The experimental results for various attentional position configurations are presented in Fig. 13. The optimal performance is achieved when the attention mechanism is positioned between the second and third convolutional layers. In this configuration, both OA and AA exhibit consistent, though modest, improvements across all datasets, with the Pavia dataset reaching an OA of approximately 90.00%, representing a 5%–8% increase over other configurations. The positional differences have a more pronounced impact on the Kappa statistic, which is particularly sensitive to category imbalance. Notably, in the Pavia dataset, the optimal configuration increases the Kappa value by approximately 10%, reaching around 85.00%, compared with about 75.00% in other configurations. This enhancement demonstrates the improved class differentiation ability provided by the attention mechanism at this position.

The Kappa value is notably sensitive to category imbalance, underscoring the role of the attention mechanism in bolstering feature extraction in convolutional layers. The model enhances features in stages: initially, two convolutional layers extract mid-level features, providing valuable context for the attention mechanism. This process dynamically captures nonlinear variations in cross-domain features by adjusting the receptive field. The final convolutional layer is responsible for global feature integration, focusing on comprehensive feature fusion. Applying attention at this stage risks losing critical detailed information. Therefore, introducing the attention mechanism too early or too late disrupts the balance between local detail preservation and global representation, making the optimal intermediate positioning crucial for improved cross-DA.

V. Conclusion

In this article, we propose a novel method ASAL for HSI classification. The proposed SADB integrates the attention mechanism directly into the CNN, enabling better extraction of spectral details. In addition, our category confusion minimization strategy calibrates the predictive information of the classifier and mitigates the adverse effects of overconfident predictions. We also introduce a gradient computation step using a double smoothing regularization strategy to minimize the smoothing term lost from the task, which effectively reduces the distribution difference between the source and TDs, thus improving the generalization performance of the model in the TD. Extensive experiments and analyses conducted on three cross-scene HSI datasets demonstrate that our method, ASAL, surpasses state-of-the-art approaches. In future work, we aim to refine the category confusion strategy to achieve even more accurate classification results and enhance the training strategy to improve the model's stability and generalization capabilities.

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